

GRAZZANISE, Italy

WMO: 162530

Lat: 41.061N

Lon: 14.082E

Elev: 29

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.7	33.4	21.3	16.0	38.7	25.0	19.1	40.4	25.7	51.6	23.0	51.8	5.3	40	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.1	91.9	76.0	89.8	75.6	87.8	75.2	82.0	87.1	80.0	85.7	78.3	84.2	8.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.7	159.6	85.9	78.6	148.6	84.3	76.6	138.9	82.9	45.8	88.2	43.7	86.2	41.7	85.0	90.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	18.2	15.3	25.6	97.0	2.6	2.7	23.8	99.0	22.3	100.5	20.8	102.1	19.0	104.0
			24.6	84.4	2.3	3.4	22.9	86.8	21.5	88.8	20.2	90.7	18.5	93.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	47.1	47.4	51.4	56.6	63.9	72.3	76.6	77.2	71.1	63.9	55.8	49.0
	DBStd	11.82	4.87	5.03	4.56	4.51	4.60	5.15	3.54	3.68	4.47	4.76	5.61	5.31
	HDD50	353	115	99	38	4	0	0	0	0	0	16	81	
	HDD65	2664	554	492	422	255	77	7	0	0	7	77	278	495
	CDD50	4410	26	27	81	201	431	669	826	844	632	430	191	52
	CDD65	1244	0	0	0	2	43	226	361	379	189	41	3	0
	CDH74	11436	0	0	0	27	367	2076	3631	3774	1346	209	6	0
	CDH80	4143	0	0	0	2	63	699	1433	1585	345	16	0	0
Wind	WSAvg	5.9	6.2	6.4	6.7	5.8	5.8	5.6	5.5	5.1	5.5	5.6	6.2	6.4
Precipitation	PrecAvg	35.0	3.6	2.8	3.3	3.2	1.9	1.2	0.9	1.1	3.1	4.0	5.4	4.3
	PrecMax	51.2	10.2	6.2	10.5	6.6	4.5	3.5	2.5	5.4	6.6	9.7	13.4	7.7
	PrecMin	26.7	0.5	0.5	0.0	1.1	0.2	0.1	0.0	0.1	0.7	0.4	2.2	0.3
	PrecStd	6.7	2.3	1.6	2.2	1.2	1.0	1.0	0.6	1.2	1.5	2.5	2.4	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.9	64.0	69.9	77.3	86.0	93.3	94.7	96.4	89.8	82.2	73.7	65.9
		MCWB	58.1	56.5	58.8	64.6	70.0	73.3	77.2	76.3	72.0	68.4	65.8	60.9
	2%	DB	60.1	60.8	66.0	73.1	80.8	89.6	91.6	92.9	86.2	78.7	70.1	62.3
		MCWB	55.6	54.6	58.3	63.3	67.8	73.6	76.6	76.8	73.3	67.8	63.5	57.6
	5%	DB	58.6	59.0	62.8	69.7	77.4	86.4	89.5	90.0	84.0	76.6	67.8	59.9
		MCWB	54.2	53.9	56.6	61.0	67.3	73.0	75.7	76.6	72.6	67.3	61.8	55.7
	10%	DB	55.8	57.1	60.7	66.6	75.1	84.2	87.4	87.8	81.0	73.5	64.7	58.6
		MCWB	51.8	52.2	55.3	60.4	66.2	72.6	75.3	76.5	71.0	65.8	59.8	54.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.4	59.3	62.9	68.1	74.8	80.4	85.2	85.8	80.8	74.4	68.2	61.8
		MCDB	62.0	62.6	66.7	73.3	80.4	85.6	90.6	90.7	84.8	78.1	71.7	64.7
	2%	WB	56.7	56.9	59.8	65.0	71.7	78.3	82.2	82.4	77.7	71.5	65.1	58.7
		MCDB	59.5	59.6	63.6	70.2	77.2	84.4	87.2	87.5	82.5	75.8	68.8	61.2
	5%	WB	54.8	54.8	57.8	63.1	69.1	76.4	80.0	80.2	75.1	69.3	62.8	56.9
		MCDB	57.6	58.0	61.5	68.3	74.8	82.9	85.6	86.1	80.4	73.6	66.7	59.5
	10%	WB	52.6	52.9	56.0	61.0	67.0	74.6	78.1	78.5	72.9	67.1	60.5	54.8
		MCDB	55.6	56.2	59.6	65.6	73.0	81.1	83.9	84.6	78.5	71.8	64.3	57.8
Mean Daily Temperature Range	5% DB	MDBR	15.7	17.0	17.4	19.0	20.2	20.1	20.2	21.1	19.4	18.4	16.0	15.6
		MCDBR	17.2	18.6	20.9	23.1	24.1	23.5	23.3	23.9	22.0	21.1	18.3	16.9
		MCWBR	13.6	14.0	15.3	16.1	14.7	13.1	13.7	14.4	13.5	13.8	13.4	13.1
	5% WB	MCDBR	15.3	16.2	18.8	21.0	21.8	21.0	20.2	21.5	19.8	18.6	17.1	14.8
		MCWBR	13.3	13.7	15.0	16.4	15.7	14.4	15.1	15.4	14.6	14.5	14.0	12.7
Clear-Sky Solar Irradiance	taub	0.347	0.372	0.412	0.456	0.454	0.450	0.447	0.447	0.436	0.403	0.380	0.350	
	taud	2.436	2.344	2.230	2.118	2.180	2.240	2.259	2.266	2.283	2.360	2.355	2.427	
	Ebn at Noon	253	263	264	259	263	264	263	259	253	249	236	241	
	Edh at Noon	25	32	40	48	46	43	43	41	38	31	28	24	
All-Sky Solar Radiation	RadAvg	570	825	1218	1620	1984	2269	2320	2043	1490	1039	652	517	
	RadStd	58	89	142	120	142	104	123	128	69	96	62	60	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.31	N/A	N/A	+1.93	+3.06	+3.82	N/A	-310	+551	+261
(47)	Regional (18 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-142	+236	+122

Nomenclature: See separate page